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INRAE jobs

TEMPORARY POSITION OT-28962

Temporary position research : analysis and scientific publication of data on deadwood surveys in forest

 45290 Nogent-sur-Vernisson

INRAE presentation

The French National Research Institute for Agriculture, Food, and Environment (INRAE) is a major player in research and innovation. It is a community of 12,000 people with 272 research, experimental research, and support units located in 18 regional centres throughout France. Internationally, INRAE is among the top research organisations in the agricultural and food sciences, plant and animal sciences, as well as in ecology and environmental science. It is the world's leading research organisation specialising in agriculture, food and the environment. INRAE's goal is to be a key player in the transitions necessary to address major global challenges. Faced with a growing world population, climate change, resource scarcity, and declining biodiversity, the Institute has a major role to play in building solutions and supporting the necessary acceleration of agricultural, food and environmental transitions.

Work environment, missions and activities

Work environment

You will be located at the INRAE center in Nogent-sur-Vernisson within the Forest Ecosystems Research Unit (UR EFNO), which comprises approximately 40 researchers, engineers, and technicians. Research and expertise activities within the Biodiversity research domain focus, in particular, on the pressures affecting forest biodiversity, with the aim of proposing monitoring methods and recommendations for sustainable forest management and planning.

Context

As part of this Biodiversity research domain, you will contribute to the unit's work on methods and protocols for measuring forest biodiversity and the indirect indicators of it. Your analysis will focus on metrics characterizing deadwood—whose volume is considered an indirect indicator of biodiversity—and the protocols used to measure it. In 2021, approximately fifty field workers conducted repeated deadwood surveys using a standardized protocol across five clusters of six forest plots—comprising coniferous and deciduous stands—in the Domaine des Barres, following the approach previously used for other dendrometric measurements. These data have not yet been analyzed due to the unit's unavailability of permanent staff and the lack of ongoing projects related to this topic.

Objectives

We are looking for a researcher for:

(i) Main objective: to analyze and publish this dataset, particularly with regard to inter-observer variability in deadwood surveys at different scales. This work will include compiling the literature on the subject, curating the data through a second collective data entry, preparing the dataset for analysis, defining, writing, and estimating the statistical model to assess the variability in deadwood measurements at different scales—particularly between observers—and interpreting the results, including in light of the state of the art. This objective will culminate in the publication of at least one scientific peer-reviewed article presenting these results. Initially, we plan to use Bayesian models that would extend those developed by Gosselin & Larrieu (2020), i.e., latent variable models. One of the challenges will be to model the variability in different deadwood metrics (total volume, volume by type, total diversity, etc.). One approach would be to model these different metrics within the same statistical model, as outcomes of latent variables corresponding, for example, to the number and average volume of deadwood pieces per category. Systematic variations between observers will be addressed as random effects that can be structured by different characteristics of their surveys, such as observer quality factors. This work will be used in the final part of a PhD thesis, which aims to account for uncertainty in deadwood measurements when estimating the relationship between deadwood measurements and saproxylic biodiversity. If Bayesian models are used, the research fellow will benefit from the team's experience with these techniques (the runMCMCbtadjust package, coupled in particular with nimble). Other approaches are nevertheless possible (TMB, for example).

(ii) Secondary methodological objective: other applications may also be possible, particularly in the context of methodological developments that may be required for these models or in relation to applied statistical techniques (particularly Bayesian methods).

The researcher will be supervised by Dr. Frédéric Gosselin, Dr. Christophe Bouget, and Arthur Rossignol.

Please submit a cover letter, a resume, and a letter of recommendation—or at the very least, the contact information of a reference

Work at a computer workstation with access to remote servers. Little travel and little fieldwork—primarily fieldwork to put the analyzed data into context.

Training and skills

PhD

Recommended background : a Ph.D. with a strong focus on applied parametric statistical modeling, along with a solid track record of academic publications. A background in mathematics would be a plus.

Desired qualifications : applied statistics; knowledge of ecology or forestry is welcome; excellent written English skills.

Relevant experience : scientific writing; statistics using R; parametric models with latent variables.

Desired skills : strong writing skills; project management; interest in applied statistics and scientific publishing.

INRAE's life quality

By joining our teams, you benefit from (depending on the type of contract and its duration):

- up to 30 days of annual leave + 15 days "Reduction of Working Time" (for a full time);
- [parenting support](#): CESU childcare, leisure services;
- skills development systems: [training](#), [career advise](#);
- [social support](#): advice and listening, social assistance and loans;
- [holiday and leisure services](#): holiday vouchers, accommodation at preferential rates;
- [sports and cultural activities](#);
- collective catering.

How to apply

All persons employed by or hosted at INRAE, a public research establishment, are subject to the Civil Service Code, particularly with regard to the obligation of neutrality and respect for the principle of secularism. In carrying out their functions, whether or not they are in contact with the public, they must not express their religious, philosophical or political convictions through their behaviour or by what they wear. > Find out more: fonction publique.gouv.fr website (in French)

OFFER REFERENCE

- **Contract:** Temporary position
- **Duration:** 13 months
- **Beginning:** 01/09/2026
- **Remuneration:** Between 3 100 and 3 500 € gross monthly salary depending on experience
- **Reference:** OT-28962
- **Deadline:** 15/06/2026



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LIVING IN FRANCE AND WORKING AT INRAE

Our guide for international scientists

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